

X-Random Forest: An Explainable AI Framework for Early Stage Diabetes Prediction with Hyperparameter Tuning and Cross-validation

Abstract—Diabetes currently affects over 422 million people worldwide, presenting a significant global health challenge. While existing methods allow for the detection of diabetes through blood sugar levels, these methods often require expert analysis and fail to provide insights into the underlying causes and contributing factors. This limitation can result in missed opportunities for preventive measures, leading to new cases of diabetes. The complexity of diabetes is further compounded by the difficulty in identifying the most critical features among a limited set of factors. This study seeks to address these challenges by employing Artificial Intelligence to identify key factors and assess their impact on diabetes onset. We propose an Explainable Random Forest (XRF) model that not only predicts diabetes status with an impressive 98.59% accuracy, thanks to hyperparameter tuning and 10-fold cross-validation, but also enhances the interpretability of the model through Explainable Artificial Intelligence (XAI). By identifying and emphasizing the most important features, the study aims to guide potential patients in taking preventive measures and help those already affected to better manage their condition.

Index Terms—Explainable Random Forest (XRF), Early Stage Diabetes Prediction, Explainable Artificial Intelligence (XAI), Hyperparameter Tuning, Cross-validation.

I. INTRODUCTION

High blood glucose levels are a common and dangerous symptom of diabetes, which is frequently brought on by obesity or insulin malfunction. Blood sugar rises with the disturbance to the metabolism of carbohydrates. Early diagnosis is critical for effective care, thus it's imperative to investigate cutting-edge techniques. In order to improve patient outcomes, preventative healthcare, and early-stage diabetes prediction, this study looks into machine learning methodologies [1].

Diabetes is a widespread chronic disease that leads to severe health complications, including damage to vital organs. According to the WHO, around 422 million people are affected globally, with the prevalence rising significantly, particularly in low- and middle-income countries. The number of affected individuals is projected to reach 490 million by 2030. In Europe alone, diabetes caused approximately 186,000 deaths in 2019. Countries such as India, Canada, and China also face high prevalence rates, with India currently having over 40 million people living with diabetes. This highlights the urgent need for improved diagnosis and management strategies [2].

Traditional diagnostic approaches for early-stage diabetes depend heavily on human expertise, which can be susceptible to errors and variability. While these methods are valuable, they may not always deliver the precision needed for early

detection. Machine learning presents a promising alternative by leveraging large datasets to enhance predictive accuracy. Although previous research has explored various algorithms, common challenges remain. In this study, we utilize the Random Forest algorithm, emphasizing improvements in cross-validation and hyperparameter tuning to refine prediction accuracy and overcome existing limitations.

RQ1: *In what ways can machine learning, combined with XAI, accurately predict Diabetes based on regular life data?*

RQ2: *How do insights from XAI help to be precautionous about Diabetes indicating key factors?*

The abovementioned questions are meticulously tackled to conclude our study with the following contributions.

- The dataset is preprocessed in a well defined manner to obtain the optimal data for training and testing.
- The X-RF model, optimized through advanced hyperparameter tuning, is used to forecast the likelihood of Diabetes. The performance of the X-RF model is compared against four other classifiers, highlighting its superiority.
- To mitigate overfitting, a 10-fold cross-validation technique is employed, ensuring a thorough analysis.
- A dual approach to explainability, using seven different XAI methods, is implemented to interpret the black-box models and identify the key factors influencing the ML algorithms' performance.
- Insights from the XAI methods guide the precautionary factors based on the most significant features, after which not only the affected people can control the status of Diabetes but also the unaffected people can be aware of.

II. LITERATURE REVIEW

The field of early-stage diabetes prediction through machine learning is burgeoning, yet substantial research remains limited. This review consolidates existing studies, evaluating their contributions and identifying areas for improvement, while also acknowledging dataset disparities relative to our research.

Kopitar et al. [1] conducted a comparative analysis between Glmnet and Random Forest (RF) for predicting fasting plasma glucose levels, revealing that RF outperformed with a lower RMSE of 0.842 compared to 0.859 and demonstrated superior variable stability. However, the interpretability of their algorithms was notably lacking. Kaur and Kumari [2] assessed various machine learning models on the Pima Indian Diabetes dataset, finding that the linear kernel SVM achieved the highest accuracy (0.89), while k-NN excelled in recall (0.90). They

observed that Boruta-based feature selection enhanced model performance, though the visualization of selected features was inadequately addressed. Yasir et al. [3] reviewed a range of machine learning methods for diabetes prediction, with Random Forest attaining 97.4% accuracy and the Crow Search Algorithm combined with Feed Forward Neural Networks achieving 99.04%. This underscores the critical role of feature selection and algorithm choice, yet their study lacked a method to visualize the most important features.

Khafaga et al. [4] reported a 98.07% accuracy using an Extreme Learning Machine (ELM) model with questionnaire data, making it particularly suitable for resource-constrained settings due to its cost-effectiveness and user-friendly design—especially pertinent during the COVID-19 pandemic. However, the explainability of the ELM model was absent. Elsayed et al. [5] emphasized the importance of early diabetes detection, with algorithms such as Support Vector Machines and Decision Trees showing strong performance. Their integration of K-Means with J48 achieved a 90.04% accuracy, highlighting the potential of combining classification algorithms with association rules. Wijyaningrum and Saragih [6] demonstrated that MLP models with one hidden layer reached a 92.66% accuracy, while those with two hidden layers improved to 93.18% on the PIMA Indian Diabetes Dataset. Further optimization with the Improved Crow Search Algorithm (ICSA) elevated accuracy to 97.69%.

Ergün et al. [7] showed that Convolutional Neural Networks (CNN) reached a remarkable 99.04% accuracy, closely followed by XGBoost at 97.89%. Nonetheless, the interpretability of these models remains a significant limitation. Zarar and Wang [8] found that ANN models achieved the highest accuracy of 98.8% on both the PIMA Indian Diabetes and Kaggle Diabetes datasets, underlining the importance of addressing data scarcity. However, the imbalance in the dataset introduced a potential bias toward the minority class. Vakil and Pachchigar [9] reported that Random Forest achieved 99% accuracy on the Early-Stage Diabetes Risk Prediction dataset, identifying “Polyuria” as a key feature. Their findings emphasize the necessity of explainable AI for enhancing model interpretability.

Abushawish and Nassif [10] observed that Random Forest attained up to 98.49% accuracy, while Support Vector Machines reached 96.4% using data from Sylhet Diabetes Hospital. Lower accuracies were reported for Naïve Bayes and k-NN, underscoring the pivotal role of feature selection. Refat et al. [11] found that Discriminant Analysis outperformed Logistic Regression, with an ROC area of 0.837 versus 0.527, highlighting the importance of exploring diverse classification methods to enhance prediction accuracy. Gündoğdu et al. [12] developed a hybrid model (MLR-RF-XG) that achieved 99.23% accuracy with minimal computational demands on the PIMA Diabetes dataset, emphasizing the value of feature selection, though broader validation is needed.

Laila et al. [13] noted that ensemble methods, particularly Random Forest, achieved 97% accuracy, outperforming AdaBoost and Bagging. Their review underscores the ongoing

challenge of achieving near-perfect accuracy and the need for innovative algorithmic approaches. Samet et al. [14] demonstrated that a hybrid model achieved 90.62% accuracy on the PIMA Diabetes dataset, surpassing individual techniques like Naive Bayes and SVM, illustrating the advantages of algorithmic synergy. Chen et al. [15] highlighted the significance of age, BMI, and biochemical markers in predicting diabetes progression, with logistic regression emerging as the best performer, evidenced by AUC values exceeding 0.7. Their findings point to the necessity of further research and validation across diverse populations.

In our research, we address the limitations identified in previous studies by balancing the dataset, employing manual hyperparameter tuning, and utilizing 10-fold cross-validation to mitigate overfitting. Most notably, the incorporation of Explainable AI (XAI) enhances the interpretability of black-box models, enabling better determination of feature importance.

III. DATASET DESCRIPTION

The Early Stage Diabetes Risk Prediction Dataset, accessible from the UCI Machine Learning Repository [16], offers vital health-related data. This dataset comprises 520 records, capturing critical variables such as age, BMI, glucose levels, and symptoms like polyuria and polydipsia. Notably, all data are presented in numerical format. A detailed overview of the dataset is provided in Table I.

TABLE I
DESCRIPTION OF THE DATASET.

No of Samples	Format	No of classes	Source
520	Numerical	2	UCI

IV. METHODOLOGY OF THE RESEARCH

The research leverages the Early Stage Diabetes Risk Prediction Dataset to train an Explainable Random Forest (X-RF) architecture. A well-defined preprocessing strategy is first implemented, followed by manual selection of optimal hyperparameters using RandomizedSearchCV. The model is then validated using a 10-fold cross-validation approach, where each fold is treated as the testing set once, allowing performance to be assessed across iterations and averaged. To establish the superiority of the X-RF, it is compared with four other classifiers using the same approach. To further interpret the model, four explainable artificial intelligence (XAI) techniques are employed, providing insights into the decision-making process. Additionally, two feature importance determination methods are applied to pinpoint the key features influencing predictions, aiming to enhance both model accuracy and the understanding of factors contributing to diabetes risk.

A. Data Preprocessing

In the preprocessing stage, several crucial steps are undertaken to prepare the data for model training. To address the class imbalance in the dataset—originally consisting of 320 samples for class ‘1’ and 200 samples for class ‘0’—the

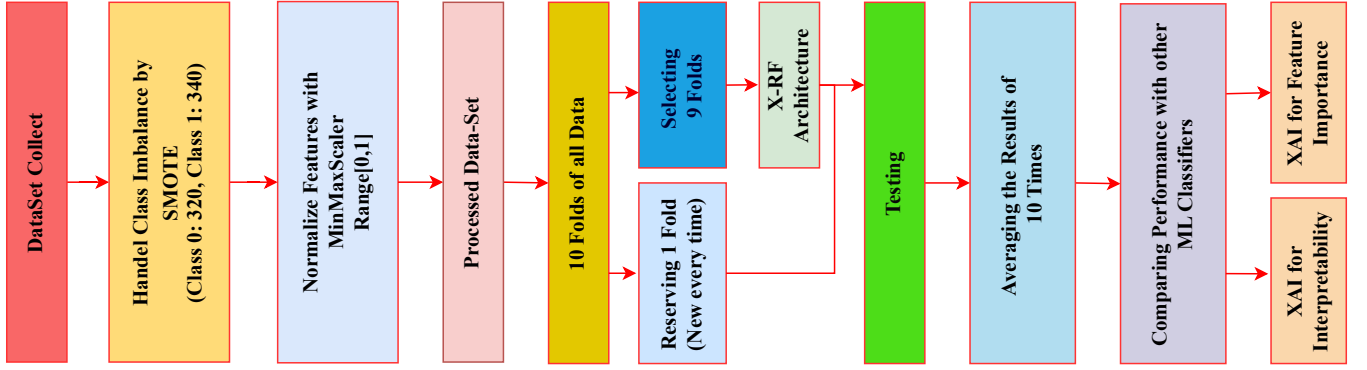


Fig. 1. Sequential Schema of Methodology.

Synthetic Minority Over-sampling Technique (SMOTE) is applied. This technique generates synthetic samples for the minority class, equalizing the class distribution to 320 samples each. Following this, the class labels are separated from the feature attributes. To ensure consistency across the dataset, feature variables are normalized to a range between 0 and 1 using the MinMaxScaler. The MinMaxScaler operates on each feature individually, applying the formula:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

Here, X represents the original feature value, $X_{\min}$ and $X_{\max}$ are the minimum and maximum values within the training set, respectively, and X' is the resulting scaled value.

B. Creation of ML Architectures

In this study, five machine learning classifiers—Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), and Extra Trees (ET)—are implemented. Optimal hyperparameters for each model are identified using RandomizedSearchCV, coupled with 10-fold cross-validation, ensuring that the most effective parameter set is selected for each classifier. The dataset is divided into 10 equal segments for cross-validation, where each model is trained on 9 segments and tested on the remaining one. This process is repeated across all folds to provide a comprehensive performance evaluation. The final performance metrics are obtained by averaging the results across all folds.

C. Utilization of Explainable Artificial Intelligence (XAI)

In our study, Explainable Artificial Intelligence (XAI) techniques are employed to enhance model interpretability and facilitate feature importance analysis. We utilize Shapley Additive Explanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), Anchors, and Integrated Gradients (IG) to provide insights into the decision-making process of the models. Additionally, Importance by Coefficients (ICE), and Permutation Feature Importance (PFI) are applied to assess the significance of individual features, revealing how input variables influence predictions and overall model performance.

V. EXPERIMENTAL RESULT ANALYSIS

A. Performance Evaluation Measures

We assess model performance by employing essential metrics such as Accuracy (Acc), Precision (Pre), Recall (Re), F1-score (F1), Specificity (Spe), Matthews Correlation Coefficient (MCC), and Cohen’s Kappa (κ). These metrics, derived from the confusion matrix, offer a holistic view of a model’s predictive effectiveness by encapsulating true positives, false positives, true negatives, and false negatives.

B. Experimental Setup

Our entire architecture was developed and executed on Google Colaboratory, utilizing the platform’s default CPU, an Intel Xeon processor with 2 virtual CPUs (vCPUs) and 12GB of RAM. For each classifier, we employed RandomizedSearchCV to identify the optimal hyperparameters, as summarized in Table II.

TABLE II
BEST PARAMETERS FOR EACH MODEL

Model	Best Parameters on Grid
RF	{‘n_estimators’: 100, ‘min_samples_split’: 10, ‘min_samples_leaf’: 2, ‘max_features’: ‘sqrt’, ‘max_depth’: 10, ‘bootstrap’: False}
SVM	{‘kernel’: ‘poly’, ‘gamma’: ‘scale’, ‘degree’: 4, ‘coef0’: 0.666, ‘C’: 3.1306}
LR	{‘solver’: ‘newton-cg’, ‘penalty’: ‘l2’, ‘max_iter’: 1000, ‘C’: 7.1457}
DT	{‘splitter’: ‘best’, ‘min_samples_split’: 2, ‘max_depth’: 12, ‘min_samples_leaf’: 2, ‘max_features’: ‘None’, ‘criterion’: ‘entropy’, ‘class_weight’: ‘balanced’}
ET	{‘n_estimators’: 1788, ‘min_samples_split’: 5, ‘min_samples_leaf’: 1, ‘max_features’: ‘sqrt’, ‘max_depth’: 70, ‘bootstrap’: False}

C. Performances of ML Architectures

In this section, we conduct a comparative analysis of the performance metrics across various classifiers to pinpoint the most effective model. Table III offers a detailed comparison of five classifiers based on critical metrics. The Random Forest (RF) classifier stands out, achieving the highest accuracy at

98.59% and an impressive F1-Score of 98.58%. With a precision of 98.76% and recall of 98.43%, RF demonstrates a well-rounded ability to correctly identify both positive and negative instances. Its high specificity of 98.75% further highlights its reliability in accurately classifying negative cases. The Extra Trees (ET) classifier is a close competitor, matching RF with an accuracy of 98.59% and an identical F1-Score of 98.59%. ET also boasts a strong recall of 98.75% and specificity of 98.59%, confirming its effectiveness. The Support Vector Machine (SVM) also delivers a solid performance with an accuracy of 96.87% and an F1-Score of 96.91%. Its precision (97.16%) and recall (96.87%) underscore its capability in balancing detection of both classes. In contrast, simpler models like Logistic Regression (LR) and Decision Tree (DT) exhibit comparatively lower accuracy and F1 scores. LR records an accuracy of 92.65% with an F1-Score of 92.70%, while DT shows an accuracy of 97.03% paired with an F1-Score of 96.94%. The superior outcomes of RF and ET are largely due to their ensemble approach, which effectively reduces variance and mitigates overfitting by aggregating predictions from multiple decision trees, resulting in more reliable predictions. SVM's strong performance is attributed to its proficiency in handling high-dimensional spaces and optimizing the decision boundary that maximizes class separation.

TABLE III
PERFORMANCE METRICS OF THE CLASSIFIERS

Model	Acc	Pre	Re	F1	Spe	MCC	κ
RF	98.59	98.76	98.43	98.58	98.75	0.972	0.971
ET	98.59	98.47	98.75	98.59	98.59	0.972	0.971
SVM	96.87	97.16	96.87	96.91	96.87	0.939	0.930
DT	97.03	98.38	95.63	96.94	98.44	0.942	0.941
LR	92.65	93.36	92.50	92.70	92.81	0.857	0.853

To further demonstrate the effectiveness of the X-RF architecture, we employ two critical evaluation tools: the Confusion Matrix and ROC-AUC analysis.

As depicted in Fig. 2, the confusion matrix for the RF model, based on 640 samples, reveals that the model correctly predicts 316 instances of class '0' and 315 instances of class '1', with only 4 instances of class '0' and 5 of class '1' misclassified. This highlights the model's high accuracy and strong ability to differentiate between classes, successfully classifying the vast majority of instances in both categories.

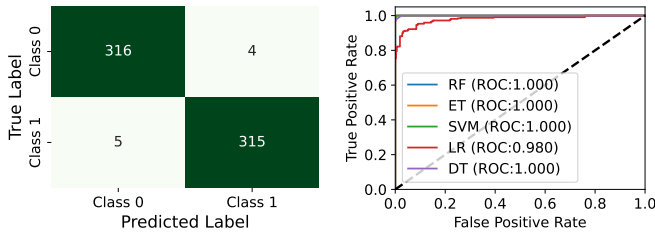


Fig. 2. Confusion Matrix by X-RF. Fig. 3. ROC-AUC for All classifiers.

In Fig. 3, the ROC curves for several classifiers are displayed. The RF, ET, SVM, and DT classifiers each achieve a

flawless AUC of 1.000, while LR performs admirably with an AUC of 0.980.

D. Explainable Artificial Intelligence (XAI)

Our study implements a two-fold explainability approach to demystify the black-box nature of the models and to identify the most influential features. This section outlines how we interpreted the proposed classifier and determined the key features contributing to its decisions.

1) **XAI for Interpreting Black-Box Models:** We employed four distinct explainability methods to interpret the inner workings of the proposed classifier, as discussed in the subsequent sections.

a) **Shapley Additive Explanations (SHAP):** This part discusses the SHAP values generated for the top-performing model. The SHAP summary plot provides insight into how each feature affects predictions across all five classifiers. Each point on the plot corresponds to a feature's SHAP value, with its color reflecting the magnitude of the feature's impact. The y-axis ranks features by importance, while the x-axis represents SHAP values. Positive SHAP values drive predictions toward class '1', while negative values steer them toward class '0'. As shown in Fig. 4, "Polyuria" stands out as the most impactful feature, followed by "Gender" and "Polydipsia". On the other hand, features like "Obesity" and "Weakness" have a relatively smaller influence, offering a clear picture of feature relevance and their respective effects on the model.

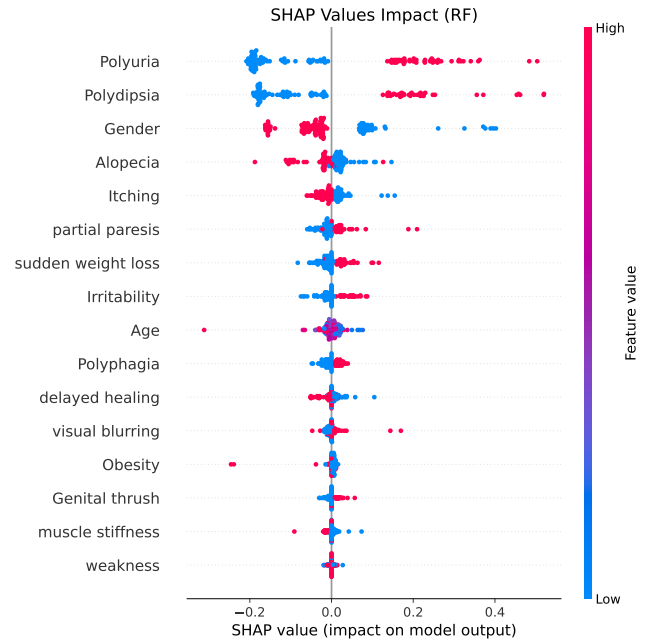


Fig. 4. SHAP Explanation (RF).

b) **Local Interpretable Model-agnostic Explanations (LIME):** This section delves into the interpretability of the Random Forest (RF) model through the use of the LIME technique. The LIME plot displays attribution scores on the X-axis, which represent how much each feature influences

the model's predictions. Bars extending to the right indicate positive contributions, which move the prediction towards class '1', while bars extending to the left represent negative contributions, pushing the prediction towards class '0'. Thanks to MinMax scaling, these scores range from 0 to 1. As depicted in Fig. 5, features such as 'Polyuria ≤ 0.00 ', 'Gender ≤ 1.00 ', and 'sudden weight loss ≤ 0.00 ' have negative contributions, steering the prediction towards class '0'. Conversely, 'Polydipsia ≤ 1.00 ' has a positive contribution, directing the prediction towards class '1'. The RF classifier achieves a high accuracy with a prediction probability of 0.98 for class '0'.

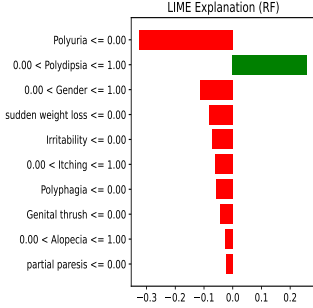


Fig. 5. LIME Explanation (SVM).

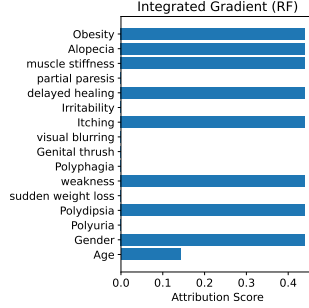


Fig. 6. Explanation by IG (RF).

c) **Integrated Gradients (IG):** This section presents the RF classifier's decision-making process as illustrated by Integrated Gradients (IG) plots. These plots display the attribution scores for each feature's influence on the model's prediction. Features are arranged vertically, and the bars indicate the degree of each feature's contribution. Positive bars reflect a positive influence, while negative bars denote a negative impact. Fig. 6 demonstrates that features such as "Obesity", "Alopecia", "muscle stiffness", "delayed healing", "Itching", "weakness", "Polydipsia", and "Gender" are the top positive contributors. Conversely, "Age" shows a moderate impact, while features like "Partial paresis", "visual blurring", "Genital thrush", "Polyphagia", "sudden weight loss", and "Polyuria" exhibit neutral scores, indicating their limited effect on predicting class '0'.

d) **anchors:** This section discusses the Anchors method for interpreting the classifier's predictions for classes '0' and '1'. The Anchors explainer utilizes class names, feature names, training data, and settings for categorical features to explain the model's decisions. The RF model, with a precision of 0.98, makes predictions based on specific feature conditions: "Polyuria" ≤ 0.00 , "Irritability" ≤ 0.00 , "sudden weight loss" ≤ 0.00 , "Genital thrush" ≤ 0.00 , "Alopecia" ($0.00 < x \leq 1.00$), and "Gender" ($0.00 < x \leq 1.00$). These conditions establish an anchor that ensures high precision in classifying the target group.

2) **XAI for Feature Importance:** Two additional methods are employed to determine the most influential features affecting the predictions of the proposed classifier.

a) **Importance by Coefficients (ICE):** Feature coefficients reveal their relative importance in the classifiers, with

larger coefficients indicating a more significant influence on predictions. Fig. 7 illustrates the coefficient values for an RF model. Features with high coefficients tend to drive predictions towards class '1', while those with low coefficients influence predictions towards class '0'. Features such as "Itching", "weakness", "Obesity", and "muscle stiffness" exhibit lower coefficients, whereas "Polyuria", "Polydipsia", "Age", and "Gender" demonstrate strong positive contributions. Features like "Alopecia" and "Polyphagia" show medium coefficients, suggesting moderate or potentially negative impacts on predictions.

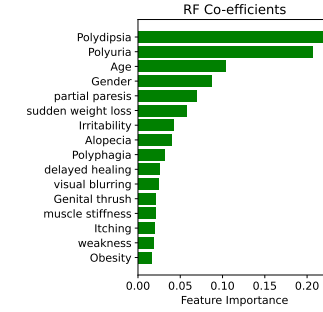


Fig. 7. ICE of RF Classifier.

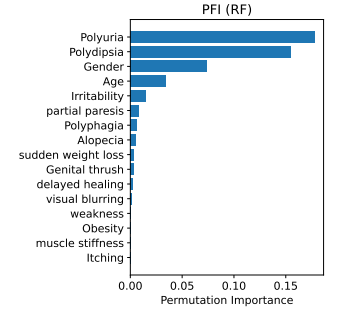


Fig. 8. PFI of RF Classifier.

b) **Permutation Feature Importance (PFI):** Permutation Feature Importance (PFI) is used to evaluate the significance of each feature by measuring the change in model performance when feature values are randomly shuffled. Fig. 8 presents the permutation importance for features in the RF classifier, displayed in a bar plot with importance values ranging from 0.00 to 0.15. Features such as "Itching", "Obesity", "Muscle stiffness", and "weakness" exhibit lower importance, while features like "Polyuria", "Polydipsia", "Gender", and "Age" stand out as the most influential. This highlights their crucial role in the model's predictions.

E. Answers to The Research Questions

Answer to RQ1: The research addresses diabetes risk prediction by utilizing an X-RF architecture. Key steps include applying a well-defined preprocessing strategy, manually selecting optimal hyperparameters with RandomizedSearchCV, and validating the model using 10-fold cross-validation. The X-RF is compared with four other classifiers to demonstrate its superiority, while four XAI techniques are employed to interpret the decision-making process. Additionally, feature importance methods are applied to identify the most influential factors contributing to diabetes risk, enhancing both accuracy and interpretability.

Answer to RQ2: Insights from XAI play a crucial role in highlighting the key factors that contribute to diabetes risk, allowing individuals to take precautionary measures. By interpreting the decisions of the X-RF model, XAI reveals which features are most significant in predicting diabetes, providing clarity on the factors that require attention. In our methodology, we apply two feature importance techniques that effectively identify these key indicators. This not only

enhances the model's transparency but also offers actionable insights, empowering individuals and healthcare providers to focus on the most relevant lifestyle habits, medical history, or other critical factors for early prevention and risk management of diabetes.

VI. DISCUSSION AND EXTENDED COMPARISON

This study assesses the performance of five machine learning classifiers for early-stage diabetes prediction using the Early Stage Diabetes Risk Prediction Dataset. Among them, the X-RF model stands out as the most effective, achieving an impressive accuracy of 98.59% and an F1-Score of 98.58%. The X-RF model excels in both precision and recall, underscoring its capability to manage complex, non-linear relationships and address class imbalance effectively. The use of various XAI techniques enhances the interpretability of the model, further supported by the evaluation of feature importance.

Overall, the X-RF architecture's exceptional accuracy and strong interpretability establish it as a leading choice for early-stage diabetes prediction. To substantiate the superiority of our approach, Table IV presents a thorough comparison with existing literature, focusing exclusively on studies that employed the same dataset as ours.

TABLE IV
COMPARISON OF OUR PROPOSED MODEL WITH EXISTING OTHERS

Article	Acc	Pre	Re	F1	Spe	MCC	κ
[1]	85.90	-	-	-	-	-	-
[2]	89.00	88.00	90.00	88.00	-	-	-
[3]	97.40	96.45	95.81	-	-	-	-
[4]	97.36	97.44	99.87	-	95.94	-	-
[5]	98.07	96.00	100	98.00	-	-	-
[6]	85.90	-	-	-	-	-	90.04
[10]	97.40	-	-	-	-	-	-
[11]	83.00	-	-	-	-	-	-
[13]	98.00	97.10	97.10	-	-	-	-
[14]	90.62	-	-	-	-	-	-
[15]	76.00	75.00	76.00	50.00	-	-	-
Ours	98.59	98.76	98.43	98.58	98.75	0.972	0.971

VII. CONCLUSION AND FUTURE WORK

This study demonstrates the effectiveness of using machine learning, particularly the Explainable Random Forest (X-RF) model, for early-stage diabetes prediction. By evaluating five classifiers—Random Forest (RF), Logistic Regression (LR), Support Vector Machine (SVM), Decision Tree (DT), and Extra Trees (ET)—the Random Forest model emerged as the most accurate to handle complex relationships and class imbalances with superior performance, making it highly reliable for diabetes prediction.

Moreover, the integration of Explainable Artificial Intelligence (XAI) techniques, including SHAP values, LIME, Anchors, and Integrated Gradients, provided meaningful insights into the key factors contributing to diabetes risk. The study identified "Polyuria" and "Polydipsia" as the most critical features, along with "Age" and "Gender." Feature importance methods such as ICE and PFI further validated these findings,

highlighting the strong correlations between these features and the model's predictions.

The X-RF architecture not only delivers high predictive accuracy but also enhances transparency, making it a valuable tool for guiding individuals at risk of diabetes to take preventive measures. Its robust interpretability also assists healthcare providers in understanding the underlying causes, offering an effective approach to diabetes management.

Future research will prioritize several key areas: expanding the use of diverse datasets, incorporating additional explainable AI techniques, and enhancing the model's generalizability across broader aspects

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